

Q1. What is linux?

>> Linux is an Operating system which is free open source, Unix-like operating system.

Q2. What is the difference between Hard Link and Soft Link?

>> Hard links and soft links are terms used in Linux operating systems. Both soft links and hard links point to files, but there's a key difference between them.

Understanding the difference between a hard link and a soft link. In Linux system, the data structure that does the actual storing of information is called an Inode. A hard link is simply another reference to the same data. If you change one, the other changes too. A soft link, on the other hand, only points to the file name, not the actual location of the data.

Q3. What is Kernal in Linux?

>> Kernal is a core operating system which basically performs like an interface between hardware and software. It also manages system resources and enables communication between system resources.

Q4. How do you create a user account?

>> To create a User account, we can go to our terminal and type the below command-

useradd "username" or sudo useradd "username" (this applies if permission denied).

```
provat@Gogol:/mnt/c/Users/prova$ useradd butan
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
provat@Gogol:/mnt/c/Users/prova$ sudo useradd butan
provat@Gogol:/mnt/c/Users/prova$ |
```

Q5. What is the grep command used for in linux?

>> grep command is used to search for specific text within files, such as words, phrases or patterns.

Q6. Step1- Create user pl.

>> To create a User, we will use **sudo useradd pl** command.

```
provat@Gogol:/mnt/c/Users/prova$ sudo useradd pl
provat@Gogol:/mnt/c/Users/prova$ sudo su pl
$ whoami
pl
$ |
```

Step2- He should be part of 3 groups g1,g2,g3.

>> To make this, first we have to create groups for that. So, creating groups in Linux is as follows.

To create a group in Linux, We will use **groupadd** command followed by the group name to create groups.

`sudo groupadd g1`

`sudo groupadd g2`

`sudo groupadd g3`

```
provat@Gogol:/mnt/c/Users/prova$ sudo groupadd g1
provat@Gogol:/mnt/c/Users/prova$ sudo groupadd g2
provat@Gogol:/mnt/c/Users/prova$ sudo groupadd g3
```

Once we done with the new groups, we will add our user(pl) to the groups.

So, for adding the user in the group, we will use **usermod** command to modify useraccount.

`sudo usermod -aG g1,g2,g3 pl`

```
provat@Gogol:/mnt/c/Users/prova$ sudo usermod -aG g1,g2,g3 pl
provat@Gogol:/mnt/c/Users/prova$ groups pl
pl : pl g1 g2 g3
```

here, -a flag- append the user to the group and -G flag specifies the list of supplementary groups.

Step3- : Whenever he creates a file automatically in the group section of file grp g1 should come.

>> To make this, we need to make group g1 as the Primary group.

For that, we need to run **`sudo usermod -g g1 pl`** command to make g1 as the primary group.

Q7. Create directory /tmp/bg as root user and create files inside it.

>> First we will go into the root user.

Command- **`sudo su`**

To create directory-

Command- **`mkdir /tmp/bg`**

```
root@Gogol:/tmp/bg# pwd
/tmp/bg
root@Gogol:/tmp/bg# |
```

To create files inside directory-

Command- **`touch test.txt`**

To print something inside the file-

Command- **`echo "Hello world!" >> test.txt`**

To print the content of the file in terminal-

Command- **`cat test.txt`**

To check the file comes as a root user or not-

Command- **`ls -l /tmp/pg`**

```
root@Gogol:/tmp/bg# touch test.txt
root@Gogol:/tmp/bg# echo "Hello world!" >> test.txt
root@Gogol:/tmp/bg# cat test.txt
Hello world!
root@Gogol:/tmp/bg# ls -l /tmp/bg
total 4
-rw-r--r-- 1 root root 13 Apr 10 18:13 test.txt
root@Gogol:/tmp/bg# |
```

